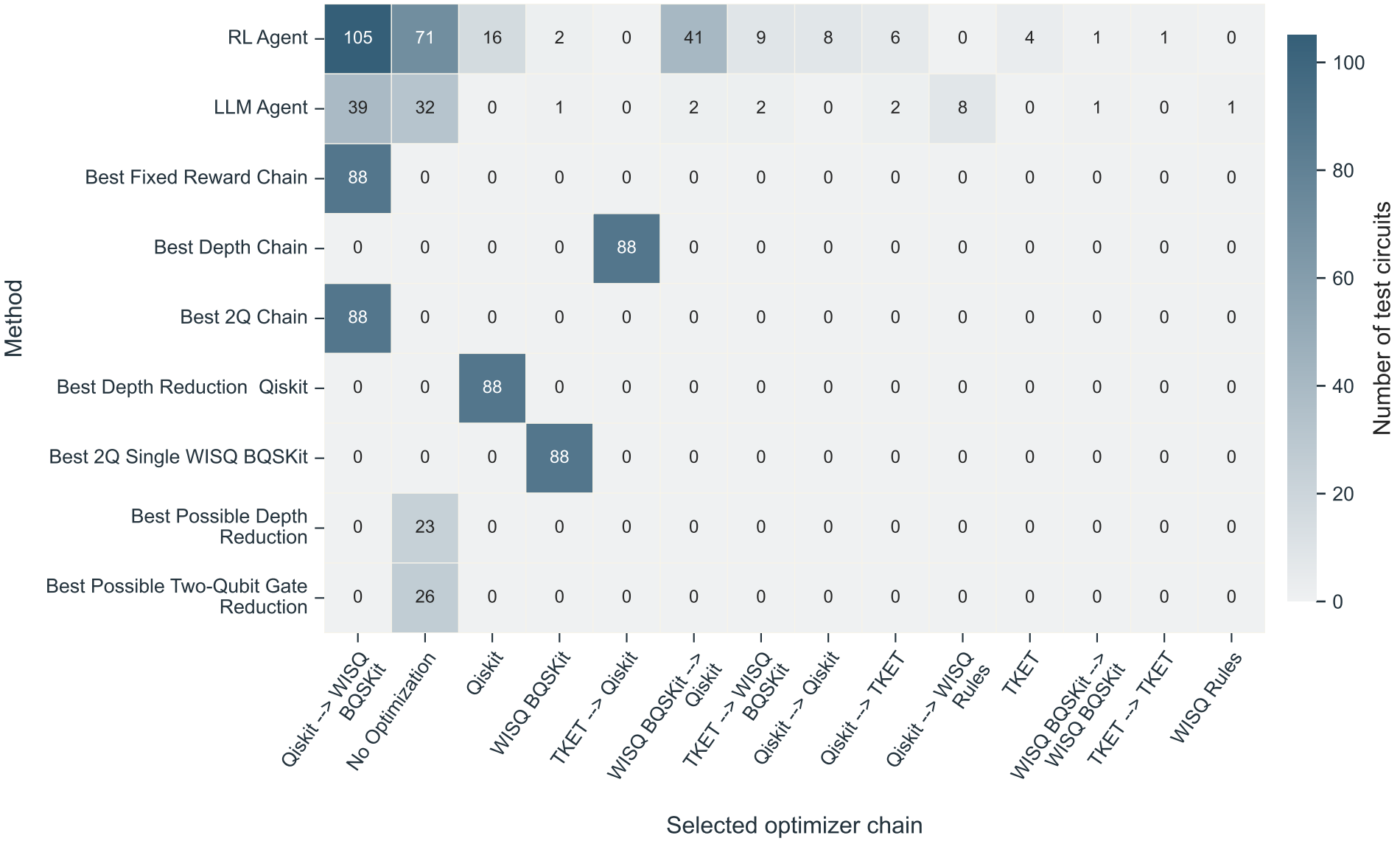
Optimizer-Chain Selection Frequency by Method
Method
RL Agent
LLM Agent
Best Fixed Reward Chain
Best Depth Chain
Best 2Q Chain
Best Depth Reduction Qiskit
Best 2Q Single WISQ BQSKit
Best Possible Depth Reduction
Best Possible Two-Qubit Gate Reduction
105 71 16 2 0 41 9 8 6 0 4 1 1 0
39 32 0 1 0 2 2 0 2 8 0 1 0 1
88 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 88 0 0 0 0 0 0 0 0 0
88 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 88 0 0 0 0 0 0 0 0 0 0 0
0 0 0 88 0 0 0 0 0 0 0 0 0 0
0 23 0 0 0 0 0 0 0 0 0 0 0 0
0 26 0 0 0 0 0 0 0 0 0 0 0 0
Qiskit --> WISQ BQSKit
No Optimization
Qiskit
WISQ BQSKit
TKET --> Qiskit
WISQ BQSKit --> Qiskit
TKET --> WISQ BQSKit
Qiskit --> Qiskit
Qiskit --> TKET
Qiskit --> WISQ Rules
TKET
WISQ BQSKit --> WISQ BQSKit
TKET --> TKET
WISQ Rules
Selected optimizer chain
Number of test circuits
100
80
60
40
20
0